

Classification System for Fence/Buried Perimeters, Pipelines, and Data Networks



HIGHLIGHTS

Applications

- Full classification system for long range security of fence and buried perimeters, wall tops, pipelines, pipeline leaks, data cable networks, vehicle and train detection
- Uses AI and pattern recognition to detect and classify: human footsteps walking normally or running, animal detection, manual digging, machine digging, engine or heavy machinery, pipeline DAS leak, tunneling, fence climb, fence cut, vehicle and train detection
- Range per controller 10 m to 100 km
- Uses fiber-optic sensors, immune to EMI/RFI/Lightning
- Designed for high security

Intelligent Redundant System

- First 100% intelligent redundant controller by design
- Eliminates the single point of failure of controller
- 2nd system automatically turns on when 1st system fails
- Hot-swappable redundant controller
- Automatic data synchronization technology
- Complete redundant system in only 4U size

High Reliability

- The most reliable system by design
- Critical data is never lost with built-in memory backup
- Redundant hard drives and power supply

Top Performance

- AI Classification algorithms for high PD and low NAR
- Typical +/- 6 m location accuracy
- Cut immunity

Easy Deployment

- Easy system installation and tuning
- No electrical power in the field
- Software defined zones and Remote troubleshooting

Flexible Integration

- Remote User Interface for monitoring multiple systems
- I/O options with XML, ASCII, modbus, gRPC, HTTP, and relays. 3rd party integration via SDK/API, GPS Mapping

DESCRIPTION

The FiberRanger FR302 is a classification system that provides security for fence/buried perimeters, pipelines, and data networks using state-of-the-art artificial intelligence and pattern recognition algorithms. Specifically, the system detects and classifies human footsteps, vehicle, manual digging, machine digging, engine, DAS leak, fence climb, and fence cuts.

The FR302 system features the most advanced Distributed Acoustic Sensor (DAS) technology based on fiber-optic sensor for perimeter security. The FR302 pinpoints the location of an intrusion attempt to within a few meters and detects multiple simultaneous events, such as coordinated intrusions. The system reports the precise location of an attempted breach using XML/ASCII or relay interfaces, which can activate PTZ cameras for verification and a response. The precise location-sensing ability, advanced artificial intelligence and classification algorithms enable FR302 to have a high probability of detection and very low nuisance alarms in the industry.

The FR302 is the first system to offer both cut immunity and 100% intelligent hardware redundancy. When a fence sensor is cut in a cut-immune configuration, the FR302 system remains fully functional and immediately reports the incident. When the controllers are installed in a redundant hardware configuration, as soon as the primary controller fails, the secondary controller automatically turns on and start protecting the perimeter. Any changes to the primary get immediately transferred to the secondary. Both cut immune and 100% intelligent redundancy architecture make FR302 one of the most reliable and compact solution available to the market for high security.

SPECIFICATIONS

System

Max Sensor Length	100 km per system, 50 km per sensor, 2 sensors per system
Sensor Installation	Straight line or loop for cut immunity
Location Accuracy	+/- 5 m (16 ft) typical
Installation Medium	Fence, buried, or hybrid
Max Virtual Zone	No limit (< 50 per km recommended)
Performance	Pattern recognition algorithms for classification, and for high Probability of Detection and low Nuisance Alarm Rate
Class and Typical Detection Range	Depends on soil type, environmental noise, intrusion type and intensity
Human walking	2-10 m (6 – 33 ft)
Human Running	10-15 m (33 – 49 ft)
Vehicle	10-20 m (33 – 66 ft)
Train	50-200 m (165 – 660 ft)
Manual Digging	10-20 m (33 – 66 ft)
Machine Digging	10-25 m (33 – 82 ft)
Engine	10-25 m (33 – 82 ft)
Tunneling	10 – 20 m (33 – 66 ft)
Pipeline DAS Leak	Depends on the leak
Cable Burial Depth	0.3 – 1.5 m (1 - 5 ft)

Features for High Reliability

Cut Immunity	Sensors can be installed in a loop to keep the system protecting with a cable cut
Redundant hardware Configuration (optional)	First Intelligent redundant configuration by design to provide 100% controller redundancy
Power Supply	Dual redundant power supplies
Hard Drive	Redundant raided hard drives

Sensor Cable

Cable Type	Gel-filled loose tubes, armored or unarmored, Outdoor rated cable
Optical Loss (Installed)	< 0.25 dB/km (0.40 dB/mi) @ 1550nm
Sensing Fiber Type	Corning SMF-28e+ or equivalent
Cable Outer Diameter	Typical 0.44" (1.1 cm)
Cable Weight	55 lbs/kft (82 kg/km)
Optical Connectors	SC/APC

Electrical

Consumption	120 Watts maximum
Requirements	100/240 Volts, 50 / 60 Hz
Consumption / Field:	None

Interfaces

Default Alarm Interface	XML, ASCII, Modbus, HTTP, gRPC over TCP/IP
Optional Alarm Interface	Relay contacts (outputs and/or inputs) D102a

Mechanical

Dimensions	19" x 20" x 3.5" (48 x 51 x 9 cm)
Rack Space	2 RU (3.5")
Weight	25 lbs (11.4 kg)

Environmental

Temperature of Controller	0°C to 50°C (32° F to 122°F)
Temperature of Cable (in Field):	-40°C to 70°C (-40° F to 158°F)
Humidity of Controller	20% to 80%, non-condensing
Humidity of Cable (in Field)	No restriction

Certification and Compliance

Electromagnetic Compatibility	FCC Part 15 Class A EC EMC Directive 2004/108/EC
-------------------------------	---



Other Related Products

FR202-Hybrid	Up to 20km point locating for hybrid installations
FR101-Zone	2 Zones (up to 1km zone length, 20 km insensitive lead)

FiberRanger LLC
Email: Sales@FiberRanger.com
Phone: +1 (267)-755-8600
www.FiberRanger.com
Pennsylvania, U.S.A